



**ROYAL SCHOOL OF APPLIED & PURE SCIENCES
(RSAPS)**

DEPARTMENT OF PHYSICS

**COURSE STRUCTURE & SYLLABUS
(BASED ON NATIONAL EDUCATION POLICY 2020)
FOR**

**M.Sc. IN PHYSICS
(1 YEAR)**

**W.E.F
AY - 2026 – 27**

STRUCTURE OF THE SYLLABUS FOR 1 YEAR PG PROGRAMME

SCHOOL NAME - ROYAL SCHOOL OF APPLIED AND PURE SCIENCES

DEPARTMENT NAME - PHYSICS

PROGRAMME NAME - 1 YEAR M.Sc. in PHYSICS

ELIGIBILITY: 4 YEAR B.SC. IN PHYSICS (HONOURS/HONOURS WITH RESEARCH)

1st SEMESTER				
COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
PHY014C101	Condensed Matter Physics	500	4	4-0-0
PHY014C102	Quantum Mechanics	500	4	4-0-0
PHY014C103	Atomic and Molecular Physics	500	4	4-0-0
PHY014C104	Electrodynamics & Electronics	500	4	4-0-0
PHY014C115	Advanced Physics Lab	500	4	0-0-8
PHY014Z191	Swayam 1	500	3/4/5	
TOTAL CREDIT FOR 1st SEMESTER			20 + 3/4/5	
2nd SEMESTER				
COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
PHY014C201	Statistical Mechanics	500	4	4-0-0
PHY014C202	Nonlinear Optics and photonics	500	4	4-0-0
PHY014C203	Astrophysics	500	4	4-0-0
PHY014C204	Physics of Nanomaterials	500	4	4-0-0
PHY014C221	Project	500	10	
PHY014Z291	Swayam 2	500	3/4/5	
TOTAL CREDIT FOR 2nd SEMESTER			26 + 3/4/5	
OR 2nd SEMESTER (For students in 2nd semester with Research)				
PHY014C222	RESEARCH PROJECT	500	26 + 3/4/5	0-0-0
TOTAL CREDIT FOR 1 YEAR PG = 46+ 6/8/10				

Level: Semester I**Course Level: 500****Name of the Subject: Condensed Matter Physics****Subject Code: PHY014C101****Scheme of Evaluation: Theory****L-T-P: 4-0-0****Total credits: 4**

Course Objectives: On completion of this course, students will be able to gain interest on the following: Electrons' motion in one-dimensional potential well, three-dimensional potential well, quantum state and degeneracy, Bloch function, Kronig-Penney model, number of states in a band, Energy gap etc.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the Free electron theory of metals, Band theory of solids, Magnetic properties of solids etc.	BT 1
CO 2	Understand electrons motion in one-dimensional potential well, insulators and semiconductors etc.	BT 2
CO 3	Apply the Concepts of quantum theory of diamagnetism and Para magnetism, diamagnetic and paramagnetic susceptibilities	BT 3
CO 4	Analyze the Critical temperature-persistent current-occurrence of super conductivity	BT 4

COURSE OUTLINE:

Modules	Topics / Course content	Periods
I	Free electron theory of metals: Free electron model, Electrons moving in one dimensional potential well, three-dimensional potential well, quantum state and degeneracy, the density of states, Fermi-Dirac statistics, effect of temperature on Fermi distribution function, the electronic specific heat. Electrical conductivity of metals, relaxation time and mean free path, electrical conductivity and Ohm's law, Thermal conductivity, Wiedemann - Franz law, thermionic emission, Hall effect.	15
II	Band theory of solids: Elementary ideas of formation of energy bands. Bloch function. Kronig-Penney model, number of states in a band, Energy gap. Distinction between metals, insulators and semiconductors. Concept of holes, equation of motion for electrons and holes, effective mass of electrons and holes. Band structure of semi-conductors, Intrinsic and extrinsic semiconductors, expression for carrier concentration (only for intrinsic), ionization energies, charge neutrality equation.	15
III	Magnetic properties of solids: Fundamental concepts, quantum theory of diamagnetism and Para magnetism, diamagnetic and paramagnetic susceptibilities of free electrons, molecular field theory of ferromagnetism, anti-ferromagnetism and ferrimagnetism, anisotropic energy, electron paramagnetic resonance and nuclear magnetic resonance, Bloch equations, Heisenberg Hamiltonian for exchange interaction.	15
IV	Superconductors: Critical temperature-persistent current-occurrence of super conductivity, ideal and non-ideal superconductors-Destruction of super conductivity by magnetic field Meissner effect- heat capacity-energy gap- microwave and infrared properties- Isotope effect- BCS theory (qualitative)- Josephson tunneling-exotic superconductors- high Tc superconductors.	15
	Total	60

Text books:

1. Solid State Physics; C. Kittel, John Wiley & sons, Inc., 8th Ed., 2005, New Delhi.
2. Solid State Physics Dekker A. J. Macmillan India Ltd, 1st Ed., 2000, Bangalore.

Reference books:

1. *Verma A.R. and Srivastava O.N.; Crystallography Applied to Solid State Physics, 2nd edition, New Age International Publishers, 2001, Darya Ganj, Delhi*
2. Gupta H.C., Solid State Physics, Vikas Publishing House, 2nd Ed., 2010, New Delhi

NPTEL LINK: <https://archive.nptel.ac.in/courses/115/106/115106061/>

Level: Semester I**Course Level: 500****Name of the Subject: Quantum Mechanics****Type of Course: Core****Subject Code: PHY014C102****Scheme of Evaluation: Theory****L-T-P: 4-0-0****Total credits: 4****Course Objectives:**

This course will help the students in understanding time dependent dynamics. It will also help the student to analyze the consequences of the indistinguishability of identical quantum particles, relativistic quantum mechanics etc.

Course Outcomes:

On successful completion of the course, the students will be able to:		
CO	Course Outcome	Blooms Taxonomy Level
CO 1	Remember Time dependent phenomena: Heisenberg picture for a time-dependent Hamiltonian; the sudden approximation.	BT 1
CO 2	Understand the scattering cross-section in quantum mechanics; scattering amplitude; the inhomogeneous Schrodinger equation for a two-particle scattering	BT 2
CO 3	Apply the quantum mechanical concept on Identical Particles: Consequences of the indistinguishability of identical quantum particles; the exchange operator; symmetric and antisymmetric wave functions	BT 3
CO 4	Analyze Klein-Gordon equation; problems related to its interpretation; Dirac Hamiltonian and equation for a free particle; determination of the Dirac matrices	BT 4

COURSE OUTLINE:

Modules	Topics / Course content	Periods
I.	Time dependent phenomena: Heisenberg picture for a time-dependent Hamiltonian; the sudden approximation; Time-dependent perturbation theory; perturbation expansion; first order transition; harmonic perturbation; transition into a continuous spectrum; Fermi golden rule; interaction of matter with electromagnetic radiation; electric dipole approximation; Rayleigh scattering.	15
II.	Theory of scattering: Scattering cross-section in quantum mechanics; scattering amplitude; the inhomogeneous Schrodinger equation for a two-particle scattering; integral equation for the wave function using the technique of Green's function; the integral equation for the scattering amplitude; the successive Born approximations; the scattering amplitude in the first Born approximation for a Central potential; Scattering by a Screened Coulomb potential; the Rutherford Scattering formula; validity of Born approximation; Method of partial waves; asymptotic solution, the scattering amplitude; general expression for scattering cross-section, the S-wave scattering	15
III.	Identical Particles: Consequences of the indistinguishability of identical quantum particles; the exchange operator; symmetric and antisymmetric wave functions; Pauli exclusion principle; inclusion of spin; Pauli spin matrices; spin functions for a two-electron system; the wave function for the ground state of helium atom. Particle number representation in quantum mechanics; application and creation operators; applications to fermions and bosons.	15

IV.	Relativistic Quantum Mechanics: Klein-Gordon equation; problems related to its interpretation; Dirac Hamiltonian and equation for a free particle; determination of the Dirac matrices; probability density and probability current density; existence of spin of the Dirac particle; Dirac electron in an electromagnetic field; non-relativistic reduction; spin magnetic moment of the electron; negative energy states and Dirac's hole theory; prediction of positrons.	15
Total		60

Text books:

1. *Introductory Quantum Mechanics; Liboff R.L., Pearson Education, 4th Ed., 2007, New Delhi.*

Reference Books:

1. L.I. Schiff, Bandhyopadhyay J.; Quantum Mechanics, McGraw Hill Education; 4th Ed., 2017, New Delhi
2. Ghatak A. K. and Lokanathan S. Quantum Mechanics, Laxmi Publications Pvt. Ltd, 5th Ed., 2015, New Delhi

NPTEL LINK: [NPTEL :: Physics - NOC:Advanced Quantum Mechanics with Applications](#)

Level: Semester I**Course Level: 500****Name of the Subject: Atomic and Molecular Physics****Type of Course: Core****Subject Code: PHY014C103****Scheme of Evaluation: Theory****L-T-P: 4-0-0****Total credits: 4****Course Objectives:**

This course begins with a review about the modern developments in experimental techniques, especially spectroscopy, and to realize the role and practical application of physics of atoms and molecules in the modern world. Hyperfine structure, Classical and Quantum mechanical concept, coupling scheme and idea of symmetry are the importance of this syllabus.

Course Outcomes:

On successful completion of the course, the students will be able to:		
CO	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the Quantum states of Electron in atoms, Hydrogen atom spectrum, LS–JJ coupling Schemes.	BT 1
CO 2	Understand, Hyperfine structure and isotopic shift, one and two electron systems, Lande's g factor, Selection Rules, Stark effect, Inner Shell vacancy.	BT 2
CO 3	Apply the concept of lattice relaxation times, and Zero field splitting	BT 3
CO 4	Analyze the concept of Born-Oppenheimer approximation, idea of symmetry for diatomic and polyatomic molecules	BT 4

Detailed Syllabus:

Modules	Topics / Course content	Periods
I.	Quantum states of Electron in atoms, Hydrogen atom spectrum, relativistic correction for energy levels of hydrogen atoms, Electron Spin, Stern Gerlach Experiment, Spin Orbit interaction, Two electron systems, LS–JJ coupling Schemes, Fine structure, Spectroscopic terms and selection rules.	15
II.	Hyperfine structure and isotopic shift, width of spectral lines, spectrum of helium and alkali atoms, Zeeman and Paschen Back Effect of one and two electron systems, Lande's g factor, Selection Rules, Stark effect, Inner Shell vacancy.	15
III.	Classical and Quantum mechanical description, Spin-spin and Spin-lattice relaxation times, Chemical shift. ESR, Basic principles, Nuclear interaction and Hyperfine Structure, Zero fields splitting.	15
IV.	Born-Oppenheimer approximation, rotation, vibration and electronic structure of diatomic molecules, the Franck-Condon principle, electron spin and Hund's cases, idea of symmetry for diatomic and polyatomic molecules	15
Total		60

Text books:

1. *Physics of Atoms and Molecules*, B.H. Branden and C. J. Jochain, Pearson Education, 2nd Ed., 2015, Noida
2. *Fundamentals of Molecular Spectroscopy*, C.N. Banwell, McGraw-Hill, 4th Ed., 2004, New York

Reference books:

1. H. E. White, *Introduction to Atomic Spectra*, McGraw-Hill Book Company, 1st Ed., 1934, New Delhi
2. G. Herzberg, *Molecular Spectra and Molecular Structure*, 2nd Ed., 2008, Vancouver
3. W. Demtroder, *Laser Spectroscopy*, Springer, 3rd Ed., 2003, New Delhi
4. Hollas, *Modern Spectroscopy*, Wiley India Pvt. Ltd, 4th Ed., 2010, Noida

5. Level: Semester I

Course Level: 500

Name of the Subject: Electrodynamics & Electronics

Subject Code: PHY014C104

Scheme of Evaluation: Theory

L-T-P : 4-0-0

Total credits: 4

Course Objectives:

The objective of this course is to provide students with a comprehensive understanding of Maxwell's equations, electromagnetic wave propagation, relativistic electrodynamics, waveguides, antennas, and embedded systems. The course aims to develop a strong foundation in the theoretical and practical aspects of electromagnetic fields, wave propagation in different media, antenna characteristics, and guided wave structures. It also intends to introduce students to the fundamentals of embedded systems and 8051 microcontrollers, including programming, interfacing, timer operations, and serial communication protocols. Through this course, students will acquire analytical and problem-solving skills required for applications in modern communication systems, electronics, and embedded technologies.

Course Outcomes:

Course Outcomes:

On successful completion of the course the students will be able to:		
CO	Course Outcome	Blooms Taxonomy Level
CO 1	Remember Maxwell's equations, electromagnetic wave propagation, relativistic electrodynamics, waveguides, antennas, embedded systems and microcontrollers.	BT 1
CO 2	Understand the concepts of relativistic electrodynamics, including Lorentz transformations, relativistic momentum, four-vector formulation and electromagnetic field tensors, along with the architecture and programming concepts of embedded systems and 8051 microcontrollers, including interfacing, timer programming and serial communication.	BT 2
CO 3	Apply the concepts of electrodynamics, antenna theory, wave propagation and embedded system programming to solve numerical and practical engineering problems.	BT 3
CO 4	Analyze the propagation characteristics and operating modes of rectangular and circular waveguides, dielectric slab waveguides and guided surface waves, while developing basic embedded applications using sensors, actuators, UART communication and Integrated Development Environments (IDE) for microcontroller programming.	BT 4

COURSE OUTLINE:

Modules	Topics / Course content	Periods
I	Maxwell's equations and Electromagnetic Wave Propagation: Maxwell's equations, fixing of Ampere law by Maxwell, Maxwell's equations in matter, conservation of energy, Maxwell's stress tensor and conservation of momentum in electrodynamics. Wave equation, propagation of electromagnetic waves in dispersive media and conducting medium, concept of waveguide. Gauge transformations of Maxwell's equations; Retarded and Lienard-Wiechert potentials.	15
II	Relativistic electrodynamics: Concept of proper and ordinary time, velocity, force and momentum; relativistic potential, magnetism as a relativistic phenomenon; Lorentz transformation of four vector potential, transformation of fields, field tensor; maxwell's equations from field tensor; Minkowski force on a charged particle;	15

III	Waveguides and Antennas: Rectangular waveguide, circular waveguide, dielectric slab waveguide surface guided waves, TE and TM modes, waveguide components. Antenna parameters, radiation from simple dipole and aperture, concept of antenna arrays, end fire and broadside arrays, horn antenna, microstrip antenna, parabolic disc antenna. Ground wave, space wave and ionospheric propagation. Communication link budget for ground transmission.	15
IV	Introduction to Embedded Systems and Microcontrollers: Fundamentals of Embedded Systems: Definition and characteristics, Components, Classification; Real-time and non-real-time systems; Applications in physics instrumentations. Introduction to Microcontrollers: Microprocessor vs Microcontroller; Architecture of microcontrollers; CPU, memory, input/output ports, timers, counters, ADC/DAC; Basics of popular microcontrollers: 8051 and Arduino. Embedded System Hardware: Sensors and transducers, digital and analog signals; Interfacing LEDs, switches, relays, displays, and motors, Power supply and clock circuits; Communication protocols.	15
	Total	60

Text Book:

1. Introduction to Electrodynamics; Griffiths D.J., PHI, 4th Ed., 2013, New Delhi
2. Integrated Electronics; Jacob Millman, Christos Halkias and Chetan D. Parikh, McGraw Hill Education; 2nd edition (1 July 2017).
3. Handbook of Electronics; Gupta & Kumar, Pragati Prakashan, 38th Edition 2012.
4. Shibu KV, "Introduction to Embedded Systems", Tata McGraw Hill Publication 1st edition (1 July 2017).

Reference Book:

1. Chakraborty B.; Principles of Electrodynamics, Books & Allied Ltd, 1st Ed., 2010, Kolkata
2. Chattopadhyay D.; Electronics: Fundamentals & Applications; New Age International, 1st Ed., 2010, New Delhi.

NPTEL LINK: <https://nptel.ac.in/courses/122106025>

Level: Semester I

Course Level: 500

Name of the Subject: Advanced Physics Lab -I

Type of Course: Core

Subject Code: PHY014C115

Scheme of Evaluation: Practical

L-T-P: 0-0-8

Total credits: 4

Course Objectives: This course will help the students in understanding of Stefan's constant – Black body radiation, Planck's constant and other advanced experiments.

Course Outcomes:

On successful completion of the course, the students will be able to:		
CO	Course Outcome	Blooms Taxonomy Level
CO1	Operate different physical devices.	BT 1
CO2	Work with different apparatus like interferometer, specific charge measurement device etc.	BT2
CO3	Apply different methods in different experiments.	BT3
CO4	Analyze different experimental results with error calculations.	BT4

LIST OF EXPERIMENTS

1. Stefan's constant – Black body radiation
2. To Study of the characteristics of JFET (Junction field effect transistor) in common source configuration and evaluation of: 1. AC drain resistance 2. Trans conductance 3. Amplification factor 4. Drain Resistance.
3. Determination of e/m by Thomson method.
4. Determination of wavelength of a Diode Laser by using Fabry-perot interferometer.
5. Determination of Hall co-efficient, mobility using Hall Effect Set-up (Digital).
6. Determination of dielectric constant of solids
7. To study the characteristics of the Geiger Muller (GM) counter and hence to determine its operating voltage.
8. To determine the angular diameter of the sun with the help of a Sextant.
9. To draw the BH curve of iron by using a Solenoid and to determine the energy loss due to Hysteresis.
10. Designing of a differentiator and integrator using op-amp.
11. Designing a low-pass and high pass filter using op-amp.
12. To determine the switching characteristics of a bistable multivibrator using 555 timer IC.

Text books:

1. A text book on practical physics Mazumdar K.G., Ghosh B, Sreedhar publishers, 16th Ed. 2011, Kolkata.

Reference Books:

2. Advance Practical Physics for students; Worsnop B.L. Flint H.T. Facsimile Publisher, 2nd Ed. 2019, New Delhi.

Level: Semester II**Course Level: 500****Name of the Subject: Statistical Mechanics****Subject Code: PHY014C201****Scheme of Evaluation: Theory****L-T-P: 4-0-0****Total credits: 4**

Course Objectives: The objective of this course is to provide a comprehensive understanding of the statistical foundations of thermodynamics and their application to various physical systems. The course introduces different statistical ensembles—microcanonical, canonical, and grand canonical—emphasizing the role of the partition function and energy fluctuations. The students will study strongly interacting systems through models such as the Ising and Heisenberg Hamiltonians, exploring various approximation methods and exact solutions to understand complex interactions in many-body systems.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the fundamentals of Thermodynamics and relation with statistical mechanics, The canonical ensemble and its thermodynamics, partition function	BT 1
CO 2	Understand the basic concept of statistical mechanics to describe systems containing huge numbers of particles; Ideal Bose System, Thermodynamic behavior of an ideal Fermi Gas	BT 2
CO 3	Apply the Concepts of Thermodynamics and Statistical Physics to Phase Transitions: Phenomenology, Brownian motion	BT 3
CO 4	Analyze the Strongly interacting systems: Ising model, Bragg-William's approximation.	BT 4

COURSE OUTLINE:

Modules	Topics / Course content	Periods
I	Statistical basis of thermodynamics; the microstate and the macrostate; probability; probability distribution function; postulates of equal a priori probability; phase space; Liouville theorem. Concept of ensembles, microcanonical; canonical and grand canonical ensembles; The grand canonical ensemble and significance of statistical quantities, partition function; principle of equipartition energy. Energy of Harmonic oscillator; partition function for canonical ensemble; energy fluctuations in the canonical ensemble; partition function and Thermodynamic function for grand canonical ensemble; density fluctuations in the grand canonical ensemble.	15
II	Ideal Bose System: Thermodynamic behaviour of ideal Bose gas, Bose-Einstein condensation (Experimental evidences), Liquid Helium: two fluid hydrodynamics, Second sound. Ideal Fermi System, Black body radiation, Stefan Boltzmann law. Wien's displacement law, Thermodynamic behaviour of an ideal Fermi Gas, Degenerate Fermi Gas.	15
III	Phase Transitions: Phenomenology: First and Second order phase transitions, elementary idea of critical phenomena, Universality of critical exponents. Fluctuations. Thermodynamic fluctuations. Spatial correlations in a fluid. Brownian motion: Einstein-Smoluchowski's theory.	15

IV	Strongly interacting systems: Ising model; Idea of exchange interaction; Heisenberg Hamiltonian. Ising Hamiltonian as a truncated Heisenberg Hamiltonian. Exact solution of one-dimensional Ising system. Bragg-William's approximation (Mean field theory); Bethe-Pearls approximation.	15
	Total	60

Text books:

3. R. K. Patharia; Statistical Mechanics, Academic Press, 3rd Ed., 2011, Cambridge.
4. Gupta and Kumar; *Statistical Mechanics*; Pragati Prakashan, 24th Ed., 2015, Meerut

Reference books:

3. *Fundamentals of Statistical and Thermal Physics*; F. Reif, Sarat Book House Pvt. Ltd, 1st Ed., 2009, Kolkata
4. Lokanathan S. and Gambhi R.S.; *Statistical and Thermal Physics- An introduction*, P.H.I., 1st Ed., 2008, New Delhi.

NPTEL LINK: <https://archive.nptel.ac.in/courses/104/101/104101125/>

Level: Semester II**Course Level: 500****Name of the Subject: Non-Linear Optics and Photonics****Type of Course: Core****Subject Code: PHY014C202****Scheme of Evaluation: Theory****L-T-P: 4-0-0****Total credits: 4****Course Objectives:**

This course aims to understand the significance of Nonlinear Optics and Photonics and its contribution in communication system.

Course Outcomes:

On successful completion of the course, the students will be able to:		
CO	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the concept of linear and nonlinear optics and the different effect arising out of it	BT 1
CO 2	Understand. The basics of linear and nonlinear wave propagation and frequency mixing.	BT 2
CO 3	Apply the properties of photonics in different field of technology	BT 3
CO 4	Analyze the properties and its uses and apply in future technological amplification.	BT 4

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Introduction to Nonlinear Optics: Overview of linear vs. nonlinear optics, Historical development and significance, Applications in modern technology, Maxwell's equations in nonlinear media, Second-order nonlinear effects (e.g., second-harmonic generation), Third-order nonlinear effects (e.g., third-harmonic generation, four-wave mixing), Polarization and its role in nonlinear optics, Criteria for material selection in photonics	15
II	Nonlinear Wave Propagation and Frequency Mixing: Self-focusing and self-phase modulation, Solitons and their properties, Three-wave mixing processes, Four-wave mixing processes, Optical parametric oscillation and amplification, Techniques and applications, Coherent anti-Stokes Raman spectroscopy (CARS), Medical imaging and diagnostics.	15
III	Introduction to Photonics: Nature of Light, Wave-particle duality, Electromagnetic spectrum, Photons and their properties, Absorption, emission, and scattering, Refractive index and dispersion, Nonlinear optical effects, Lasers: principles and types, Optical fibers and waveguides, Photodetectors and sensors, Telecommunications: fiber optic communication, Medical imaging: optical coherence tomography (OCT), Industrial applications: laser cutting and welding.	15
IV	Advanced Topics on Photonics: Quantum photonics, Photonic crystals, Integrated photonics, Photonics in modern technology, advancements in various fields such as healthcare, information technology, and manufacturing Nonlinear photonic crystals, Terahertz nonlinear photonics, Nonlinear effects in fiber optics	15
Total		60

Text books:

1. *Non-Linear Optics*; R.W. Boyd; Elsevier, 3rd Ed., 2008, New Delhi
2. *Nonlinear Optics and Photonics*, by Guang S. He; Oxford University Press 2014

Reference Books:

1. G.D. Baruah, *Essentials of Non-linear Optics and Lasers*, Pragati Prakashan, 4th Ed., 2016, Meerut
2. N. Bloembergen, *Non-linear Optics*, World Scientific, 4th Ed., 2002, Singapore
3. Y.R Shen, *The principles of nonlinear optics*, , Wiley-Inter science, 1st Ed., 2010, New Jersey

4. Geoffrey (New), *Introduction to Nonlinear Optics*, Cambridge University Press 2011
5. B.B. Laud *Laser and nonlinear optics*, New Age International (P) limited, 3rd Ed., 2010, New Delhi

Level: Semester II

Course Level: 500

Name of the Subject: Astrophysics

Subject Code: PHY014C203

L-T-P: 4-0-0

Scheme of Evaluation: Theory

Total credits: 4

Course Objectives:

Starting with the fundamental principles of astrophysics, the course is designed to provide an understanding of the structure, formation, and evolution of stars, galaxies, and the universe as a whole and to familiarize students with key observational techniques and instruments used in modern astronomy. It also promote the use of computational and simulation tools in understanding and modeling astrophysical systems.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO	Course Outcome	Blooms Taxonomy Level
CO 1	Remember theoretical and practical aspects of modern observational astronomy, Photometry, spectroscopy, stellar classification, detectors, and basic information of telescopes.	BT 1
CO 2	Understand the fundamental concepts of astrophysics including celestial mechanics, stellar structure, and galactic dynamics.	BT 2
CO 3	Apply knowledge of astrophysics to practical application of observational techniques.	BT 3
CO 4	Analyze and evaluate astrophysical calculations of fundamental character.	BT 4

COURSE OUTLINE:

Modules	Topics / Course content	Periods
I	Basics of Astronomy: Evolution of Astronomy, The celestial sphere, Altitude and Azimuth, Declination and hour-angle, coordinate systems and transformation equations. Concept of time — solar time and sidereal time. Magnitude scales, apparent, absolute, and instrumental magnitudes. Measuring stellar distance method parallax and other methods to determine stellar distances, HR Diagram, Color Index, Spectral Class.	15
II	Telescopes and Detectors: Different types of astronomical telescopes, Mounting of telescope, Radio Telescope, Space based telescope, Astronomical Spectrograph, Photographic Photometry, Detectors, Radiation theory: Equation of radiative transfer — concepts of flux, intensity, and temperature. Formation of emission and absorption lines, limb darkening.	15
III	Integral theorems of hydrostatic equilibrium of stars: Interstellar Medium (dust, nebula, meteoroids, etc.), Formation of stars, Evolution of stars, Compact stars (White dwarf, Neutron star, Black Hole), Degenerate stars, mass-radius relation and Chandrasekhar mass limit, Jeans criterion, Integral theorems of hydrostatic equilibrium of stars, Transport of energy inside a star, Binary stars, Nucleosynthesis – hydrogen burning (pp chain and CNO cycle), triple alpha reaction.	15

IV	Galaxies and Universe: The Milky way Galaxy, Dark Matter, Kinematics, Hubble classification scheme for external galaxies, Normal galaxies and AGNs, Unified model, Basics of X-ray astronomy, black holes, and gamma ray bursts, Hubble's law, nucleo-synthesis, Cosmic Microwave Background radiation, Elementary ideas on structure formations Principle of Equivalence. Gravity and Geometry. Metric Tensor and its properties. Curved space time. Tensor calculus: co-variant differentiation, parallel transport, Particle trajectories in Gravitational field. Einstein's Field equations and Stress-energy tensor, Schwarzschild metric.	15
	Total	60

Text books:

1. An Introduction to Astrophysics; Baidyanath Basu, Prentice Hall Publication, 2nd Ed., 2013, New Delhi
2. An Introduction to Astronomy and Astrophysics, Pankaj Jain, CRC Press; 1st edition (8 April 2015)

Reference books:

3. V.B. Bhatia; Text Book on Astronomy and Astrophysics with elements of cosmology, Narosa Publishing House, 2nd Ed., 2001, New Delhi
4. K. D. Abhayankar; Astrophysics: Stars and Galaxies, Abe Books, 1st Ed., 2002, Hyderabad

NPTEL LINK: https://onlinecourses.nptel.ac.in/noc23_ph21/preview

Level: Semester II

Course Level: 500

Name of the Subject: Physics of Nanomaterials

Subject Code: PHY014C204

Scheme of Evaluation: Theory

L-T-P: 4-0-0

Total credits: 4

The emphasis of this course is to impart the understanding of the Nanoscale and its significance, Challenges in Nanotechnology and properties of nanoparticles, Semiconducting nanoparticles, Synthesis methods and Special Nanomaterials.

Course Outcomes:

Upon successful completion of the course students will be able to:		
CO	COURSE OUTCOME (CO)	BLOOMS TAXONOMY LEVEL
CO1	Remember the fundamental understanding of nanomaterials and their properties.	BT1
CO2	Understand the concept of various characterization tools required to study the properties of nanomaterials and various synthesis methods.	BT2
CO3	Apply the knowledge to resolve the related problems.	BT3
CO4	Analyze the concepts of the physics of nanomaterials, to solve problems, with logical interpretations and critical thinking.	BT4

COURSE OUTLINE:

Modules	Topics / Course content	Periods
I.	Introduction: Nanoscale and its significance, surface to volume ratio, exciton, Quantum Confinement: Introduction of Quantum wells, wires and dots, size and dimensionality effects, Bohr excitons, Properties of nanoparticles: Structural properties, optical properties - Blue shift, Emission Stokes shift, electrical properties, mechanical properties and magnetic properties.	15
II.	Nanostructure growth and Synthesis Techniques: Homogeneous and heterogeneous nucleation, Growth of nanocrystals in solution, Top down and bottom up approaches- Chemical route, Chemical vapor deposition, Physical vapour deposition, Magnetron sputtering, Molecular beam epitaxy, Lithography.	15
III.	Characterization of Nanomaterials: Introduction, Structural Characterization - X-ray diffraction (XRD), Small angle X-ray scattering (SAXS), Scanning electron microscopy (SEM), Transmission electron microscopy (TEM), Scanning probe microscopy (SPM), Chemical Characterization- Optical spectroscopy- Absorption and transmission spectroscopy, Photoluminescence (PL), Electron spectroscopy.	15
IV.	Electron transport in semiconductors and nanostructures: Time and length scales of the electrons in solids- Electron fundamental lengths in solids, Size of a device and electron spectrum quantization, Time scales and temporal (frequency) regimes, Statistics of the electrons in solids and nanostructures, Electron transport in nanostructures, Applications of nanomaterials: Light emitting and detecting device, photo voltaic cell, gas sensor, Spintronic devices and spin field effect transistors (SPINFET).	15
Total		60

Text books:

1. *Nanostructures and Nanomaterials: Synthesis, Properties, and Applications*; G. Cao, Y. Wang, World Scientific, 2nd Ed., 2011, Singapore

Reference books:

1. Edelstein A.S., and Cammarata, R.C., *Nanomaterials: Synthesis, Properties and Applications*, Institute of Physics, 1st Ed., 2001, Bristol
2. C. P. Poole, J. F. J. Owens, *Introduction to Nanotechnology*; Wiley India ,1st Ed.,2003, New Delhi

Level: Semester II

Course Level: 500

Name of the Subject: Project

Type of Course: Core

Subject Code: PHY014C221

Scheme of Evaluation: Presentation

L-T-P:

Total credits: 10

Course Objectives:

This course will help the students in understanding of different interesting research physics.

Student will complete under the guidance with Faculty member.

